APPLICATION MODERNIZATION

On-premises applications to containers on Kubernetes, on Google Cloud or Azure

Assess the application estate, migrate it in waves with infrastructure as code and GitOps, then operate the platform

1 **Why modernise**
Common problems with on-premises applications, and what the package is. p. 2 – 3

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Six steps, 6R and complexity classes, the target architecture. p. 4 – 6

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4 **Working together**
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Common Problems with On-Premises Applications

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Business applications that run on own servers and virtual machines tend to show the same pattern over time. The questions below come up in most estates.

01 · WHY MODERNISE

Ageing hardware and licences

- Servers and OS versions near end of support
- Renewal decisions come back every few years
- Capacity is bought for the peak

Each renewal forces a new decision

Slow, manual releases

- Deployments are done by hand, often at night
- Each environment is set up differently
- A release needs several teams

Changes wait for a release window

Hard to scale

- New capacity takes weeks to provide
- Peaks are handled by over-provisioning
- No automatic scaling

Capacity does not follow demand

Knowledge with a few people

- Server set-up is not written down
- Configuration lives on the machines
- Changes depend on one or two people

Hard to change, hard to hand over

Limited visibility

- Logs are spread across servers
- Users notice failures first
- No shared view of application health

Problems are found late

A cloud plan without a path

- A data centre exit or cloud move is on the agenda
- Unclear which applications can move, and how
- No estimate of effort and cost

A decision without an inventory

WHAT CHANGES ON KUBERNETES

Same package everywhere
one container image for dev, test and prod

Releases through Git
a change is a commit, deployed by a pipeline

Scaling by rule
replicas added and removed by load

Infrastructure as code
environments can be rebuilt from Terraform

One view of health
central logs, metrics and alerts

What Application Modernization Is

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A service package that moves on-premises applications to containers on Kubernetes in the cloud: assessed first, migrated in waves, then operated.

02 · WHAT IT IS

FOUR PARTS

Assess
inventory, 6R, target architecture, plan

Migrate
containers on Kubernetes, in waves

Automate
Terraform, Helm, Argo CD

Operate
monthly platform operations

FOR WHOM

Organisations that run business applications on own servers or virtual machines and plan to move them to Google Cloud or Azure. «GÜNER DOĞRULASIN»

WHAT IT PRODUCES

Inventory and plan → Apps on Kubernetes → Platform operated

Moved, not rewritten
The focus is replatforming: applications are packaged as containers with limited code changes, such as externalised configuration. Functional rewrites and new features are not part of the package. «GÜNER DOĞRULASIN»

WHEN IT IS USEFUL «GÜNER DOĞRULASIN»

- A data centre exit, or a hardware or licence renewal
- Applications do not scale with demand
- Releases are slow and manual
- A move to Google Cloud, Azure or both is decided

WHAT IT IS NOT «GÜNER DOĞRULASIN»

- Mainframe migration
- Functional rewrite or new features
- Application licence negotiation
- Running end-user tests (UAT stays with the customer)
- Changes by third-party software vendors
- Data warehouse and ETL migration (separate packages)

Method: From Inventory to Operations

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Six steps. The first four form the assessment and end with a plan and an estimate; the migration and operations follow on that basis.

03 · METHOD

1 Inventory

Application list from the CMDB or a template, interviews with application owners, dependency map.

2 6R classification

A strategy and a complexity class for each application.

3 Target architecture

Landing zone, clusters, IaC and GitOps, security, observability.

4 Wave plan

Applications grouped into waves; effort and cost of the migration.

5 Migration

Platform foundation, pilot application, waves, cutover and hypercare.

6 Operations

Cluster operations, upgrades, monitoring, support.

A ASSESSMENT

B MIGRATION

C PLATFORM OPS

THREE WORKING RULES

Pilot first

One application goes end to end before the waves start. The patterns built for it are reused in every wave.

Everything as code

Infrastructure is defined in Terraform and releases are managed in Git through Argo CD. Environments can be rebuilt.

A way back at every cutover

Each application has a rollback plan that is tested before it is switched over. «GÜNER DOĞRULASIN»

6R Strategies and Complexity Classes

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In the assessment each application gets one strategy and one complexity class. The class sets the migration effort for that application.

04 · 6R AND COMPLEXITY

6R STRATEGIES

Replatform	Packaged as containers and run on Kubernetes, with small changes	Main focus
Refactor	Limited code changes, e.g. externalised configuration or stateless sessions; no functional rewrite «GÜNER DOĞRULASIN»	Limited
Rehost	Moved as is to virtual machines in the cloud «GÜNER DOĞRULASIN»	Interim only
Repurchase	Replaced by a SaaS or packaged product	Classified only
Retire	Switched off; no longer needed	Classified only
Retain	Stays where it is for now	Classified only

COMPLEXITY CLASSES

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Simple

- Stateless
- No database, or one
- Up to 2 integrations
- Current runtime

Medium

- Stateful, or has its own database
- 3 to 5 integrations

Complex

- Several databases
- More than 5 integrations
- Legacy runtime or OS dependency
- Licence constraints

Repurchase, Retire and Retain are recorded in the inventory; they are not carried out in this package.

Target Architecture: Kubernetes, IaC and GitOps

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Applications run as containers on GKE (Google Cloud) or AKS (Azure). The platform is built with Terraform; releases reach the cluster through Git and Argo CD.

05 · TARGET ARCHITECTURE

DELIVERY · FROM CODE TO CLUSTER

Code repository
application code and Dockerfile

CI pipeline
build, test, image; Cloud Build, GitLab, GitHub Actions or Jenkins

Artifact Registry
container images, versioned

GitOps repository
Helm charts and values per environment

Argo CD
keeps the cluster in line with Git

GOOGLE CLOUD LANDING ZONE · VPC, BUILT WITH TERRAFORM

sync

Cloud Load Balancing

GKE cluster

dev | test | prod

applications as containers, one namespace per environment

Secret Manager

Cloud Monitoring and Logging

Cloud SQL (optional)

IAM and network policies

Every change goes through Git. Infrastructure is defined in Terraform and releases are synced by Argo CD, so each change is traceable and an environment can be rebuilt. «GÜNER DOĞRULASIN»

ON AZURE

The same design with Azure services. Terraform, Helm and Argo CD stay the same.

GKE	**AKS**
Artifact Registry	**Azure Container Registry**
Cloud Build	**Azure DevOps or GitHub Actions**
Secret Manager	**Key Vault**
Cloud Monitoring	**Azure Monitor**
Cloud SQL	**Azure SQL or PostgreSQL Flexible Server**
Load Balancing	**Application Gateway**
Terraform, Helm, Argo CD	**unchanged**

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Three Service Tiers: Assessment, Migration, Platform Ops

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The assessment measures the estate and prices the migration. The migration moves applications in waves. Platform Ops runs the platform every month.

06 · SERVICE TIERS

A Assessment

fixed fee · 3–4 weeks

- Application inventory and dependencies
- 6R strategy and complexity class per application
- Target architecture on GKE or AKS
- Wave plan
- Migration effort and cost estimate

OUTPUT

Inventory · target architecture · wave plan · migration estimate · risk list

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B Migration

in waves · length set by the inventory

- Platform foundation: landing zone, clusters, IaC, GitOps
- Pilot application end to end
- Applications moved in waves
- Cutover with a tested rollback plan
- Hypercare and handover

OUTPUT

Applications running on Kubernetes · platform as code · runbooks handed over

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C Platform Ops

monthly · minimum term 12 months

- Cluster operations and version upgrades
- GitOps pipeline maintenance
- Monitoring and alerting
- Incident support by priority
- Cost follow-up and small changes

OUTPUT

Monthly report · platform kept current · changes from a monthly capacity

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Assessment first. A migration proposal is prepared on the inventory from the assessment. Without it, the effort cannot be estimated application by application.

Assessment: Work Flow and Deliverables

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Five phases over about three to four weeks. Data is collected in one of two ways, agreed at kick-off.

07 · ASSESSMENT

INDICATIVE PLAN · BAND M

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TWO WAYS TO COLLECT DATA

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Mode A · guided

The customer team fills in the inventory template and runs agreed discovery commands. Outputs are shared with Komtaş.

Mode B · read-only access

Komtaş gets read-only access to repositories, virtual machines and the CMDB and collects the data itself. No change is made.

DELIVERABLES

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- Application inventory with dependencies
- 6R strategy and complexity class per application
- Target architecture document
- Wave plan
- Migration effort, cost estimate and team plan
- Risk list

Duration by size. Small about 3 weeks, medium about 3.5 weeks, large about 4 weeks. Customer team time: about 3, 4 or 5 days. «GÜNER DOĞRULASIN»

Migration: Phases and Waves

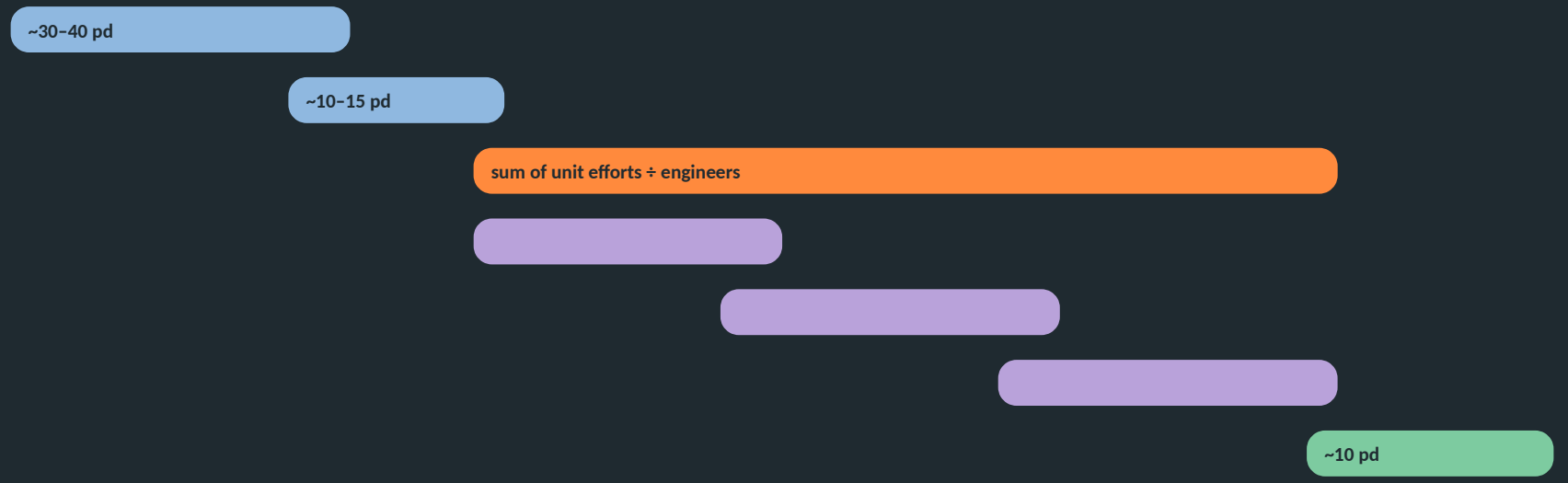
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Platform and pilot first, then the applications in waves. Phase 3 length follows from the inventory and the number of engineers.

08 · MIGRATION

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INDICATIVE · PHASE 3 LENGTH SET BY THE INVENTORY AND TEAM SIZE



PHASE 3 FORMULA

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Duration = Σ (applications per class × unit effort) ÷ engineers

UNIT EFFORT

- Simple ~5 pd
- Medium ~12 pd
- Complex ~25 pd

EACH WAVE

- Container image
- Helm chart
- Deploy via Argo CD
- Customer test
- Cutover
- Hypercare

Optional: the application's own database moved to Cloud SQL or Azure SQL. «GÜNER DOĞRULASIN»

Acceptance per application. Functional smoke test by the customer, performance on agreed scenarios with thresholds «NUMBER», rollback plan tested, running in the target through GitOps. «GÜNER DOĞRULASIN»

Pricing Approach

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How each tier is priced. Figures are given in the proposal, after the discovery call or the assessment.

09 · PRICING

A — Assessment

fixed fee

- One fixed fee, set by the size band: small, medium or large.
- The band comes from four drivers: number of applications, share of complex applications, environments, target SLA. «GÜNER DOĞRULASIN»
- Includes the migration estimate, so the next decision has a basis.

B — Migration

two options

OPTION 1 — **Unit price per application**

A price for each complexity class. The total follows the inventory; applications added later are priced the same way.

OPTION 2 — **Time and materials with a cap**

Team and rate card, with an agreed upper limit. Suits estates with many unknowns.

One option is chosen per proposal. Platform foundation, pilot and cutover are planned with fixed effort.

C — Platform Ops

monthly fee

- A monthly fee for a capacity band: small, medium or large.
- Capacity covers operations, support and small changes.
- Minimum term of 12 months. «GÜNER DOĞRULASIN»

No fixed price for the whole migration. The effort depends on what the inventory shows: dependencies, runtimes, licences. A price per application or a cap keeps the price tied to the real estate.

What the Customer Provides

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What is needed from the customer before and during the work. Most of it is access to information and the time of the people who own the applications.

ACCESS AND INFORMATION

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- **Application list**
  from the CMDB, or in the inventory template
- **Access for discovery**
  read-only access (Mode B), or discovery commands run by the customer team (Mode A)
- **Network and security input**
  connectivity to the cloud, firewall rules, security standards
- **Cloud account and approvals**
  organisation or tenant, billing account, landing zone approvals

PEOPLE AND DECISIONS

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- **A coordinator**
  single contact for access, meetings and decisions
- **Application owners**
  interviews in the assessment, decisions for each wave
- **Test team**
  user acceptance tests for each application, in the agreed window
- **Network and security team**
  connectivity, firewall and access approvals

Test capacity sets the pace. Each wave waits for the customer's tests. Planning the test team early keeps the waves on schedule. «GÜNER DOĞRULASIN»

Experience

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Container and Kubernetes migration work carried out by the Komtaş cloud team. Further references are added per proposal.

10 · EXPERIENCE



REFERENCE · ANONYMISED

An on-premises application estate moved to Kubernetes on Google Cloud

Customer name and sector are shared with the customer's permission.
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WHAT WAS DONE



Containerisation
applications packaged as container images



Kubernetes on Google Cloud
applications run on Kubernetes clusters



Terraform
infrastructure defined as code



Helm and Argo CD
releases managed through GitOps

Delivered by the cloud team. The same team carries out the assessment, the migration and the platform operations.

REFERENCES «GÜNER DOĞRULASIN»

«REFERENCE PROJECT»

«REFERENCE PROJECT»

«REFERENCE PROJECT»

What carries over. Terraform modules, Helm chart templates and pipeline templates from earlier work are the starting point; a new project adapts them. «GÜNER DOĞRULASIN»

About Komtaş, and the Next Step

Who runs Application Modernization, and what is needed to start.

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11 · ABOUT KOMTAŞ AND NEXT STEP

WHY KOMTAŞ

Google Cloud Premier partner
Komtaş is a Google Cloud Premier partner.

Microsoft Azure partner
Komtaş is also a Microsoft Azure partner; the target can be either cloud.

Kubernetes delivery experience
Applications already moved to Kubernetes on Google Cloud (p. 12).

From assessment to operations
Assessment, migration and monthly platform operations from one partner.

Since 1989, focused on data · 250+ specialists · offices in Istanbul, Ankara, Izmir and Dubai

NEXT STEPS

1. **Walkthrough** — today — you are here
2. **Discovery call** — application list, triggers, target cloud
3. **Assessment proposal** — fixed fee, by size band
4. **Kick-off** — access, inventory starts

Needed from the customer. A coordinator, a first application list, the target cloud and date, and an early start on access approvals.